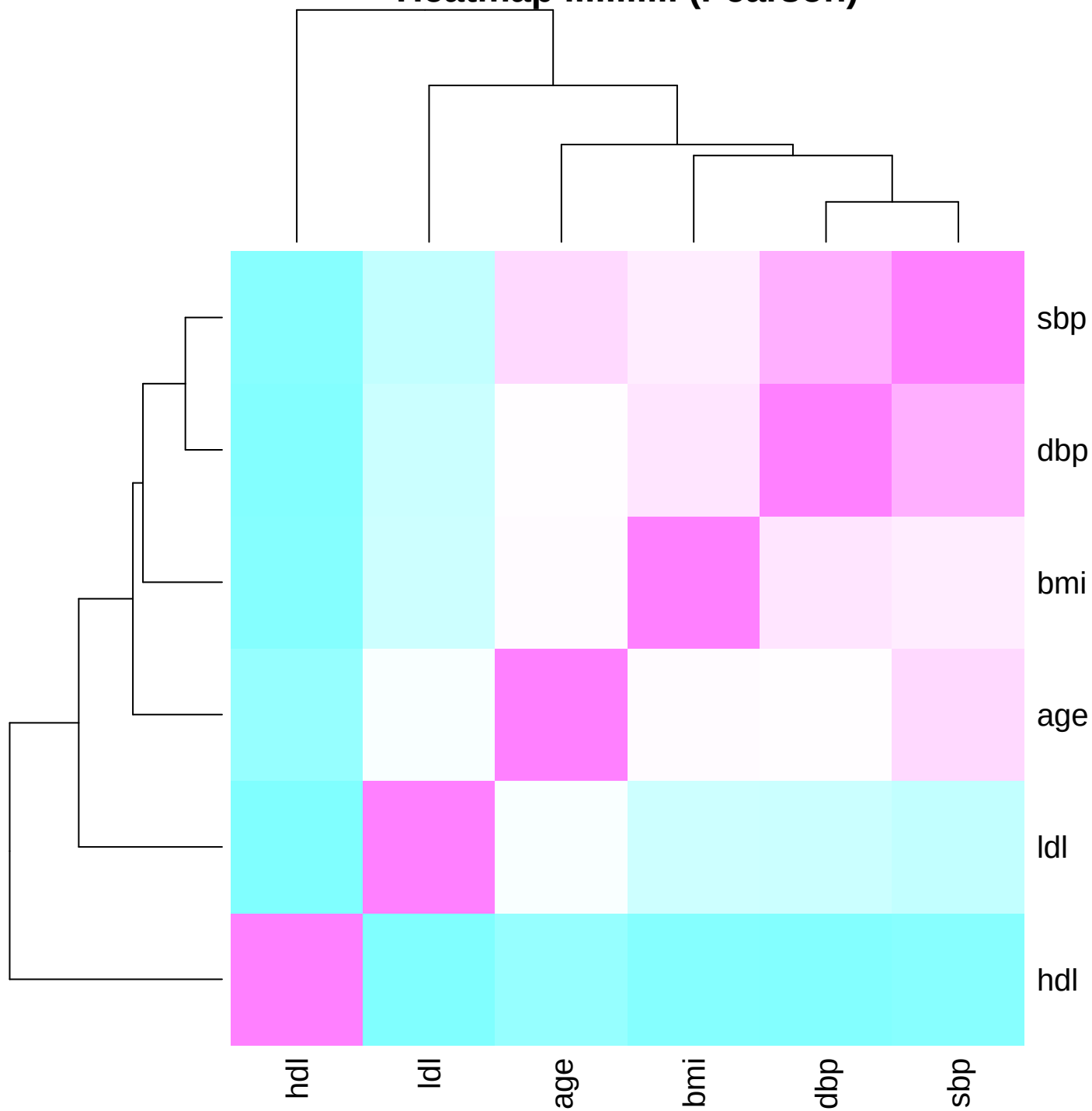


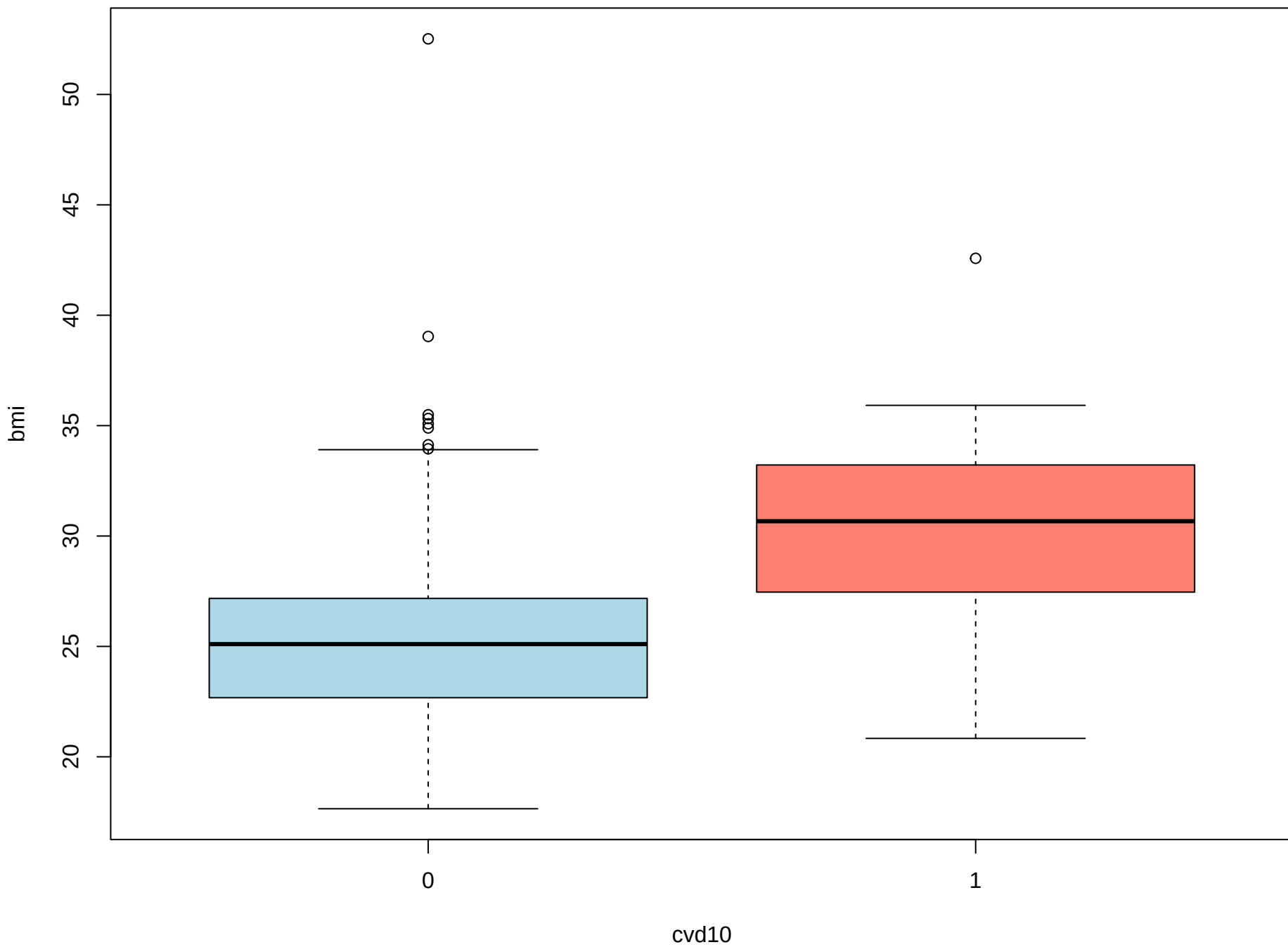
Heatmap (Pearson)



Descriptives: bmi by cvd10

	Mean	SD	Median	Skew
No_CVD	25.66	4.67	25.10	1.87
Yes_CVD	30.20	5.11	30.67	0.42

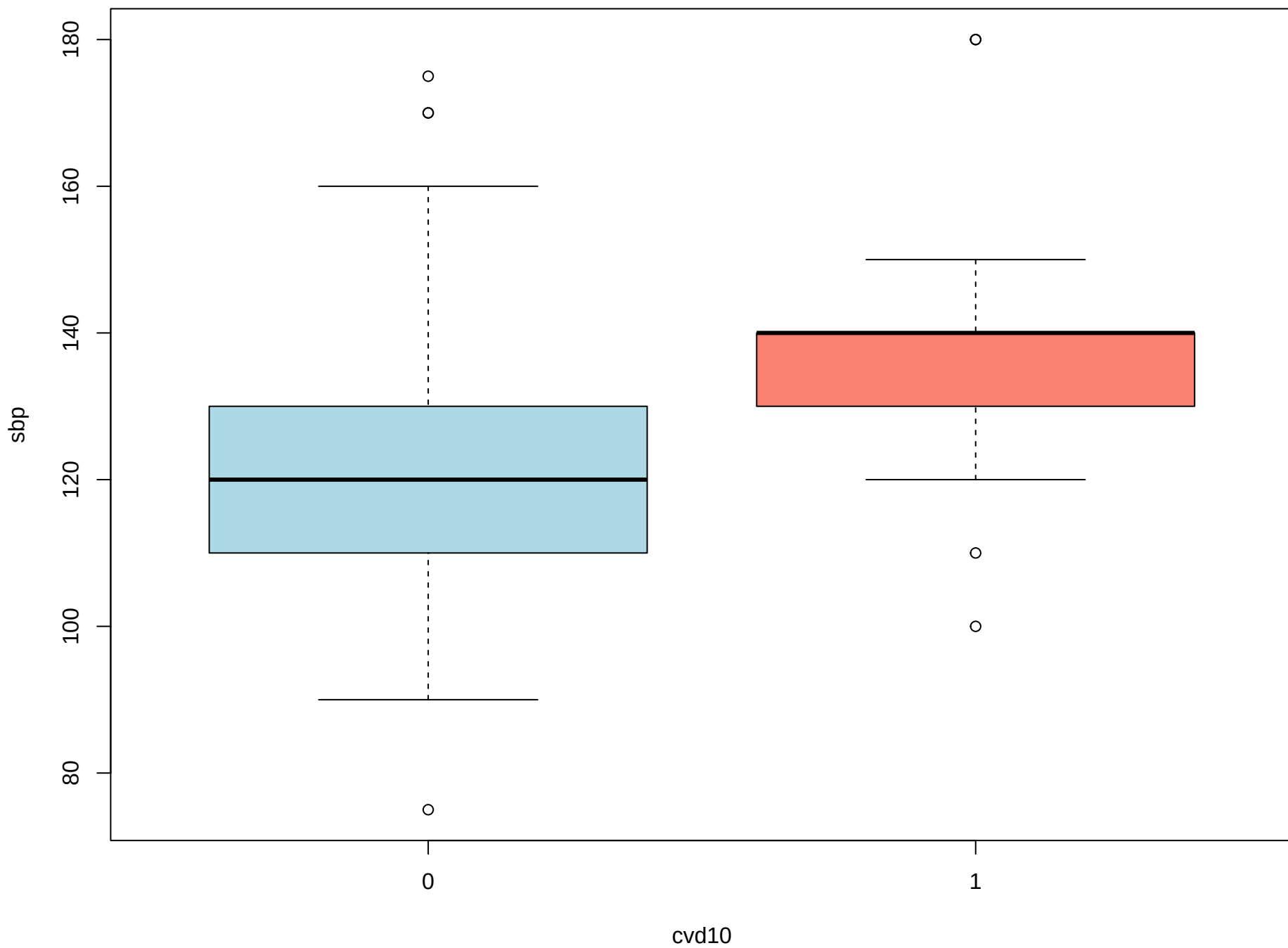
Boxplot: bmi by cvd10



Descriptives: sbp by cvd10

	Mean	SD	Median	Skew
No_CVD	120.52	18.53	120	0.58
Yes_CVD	137.06	20.62	140	0.53

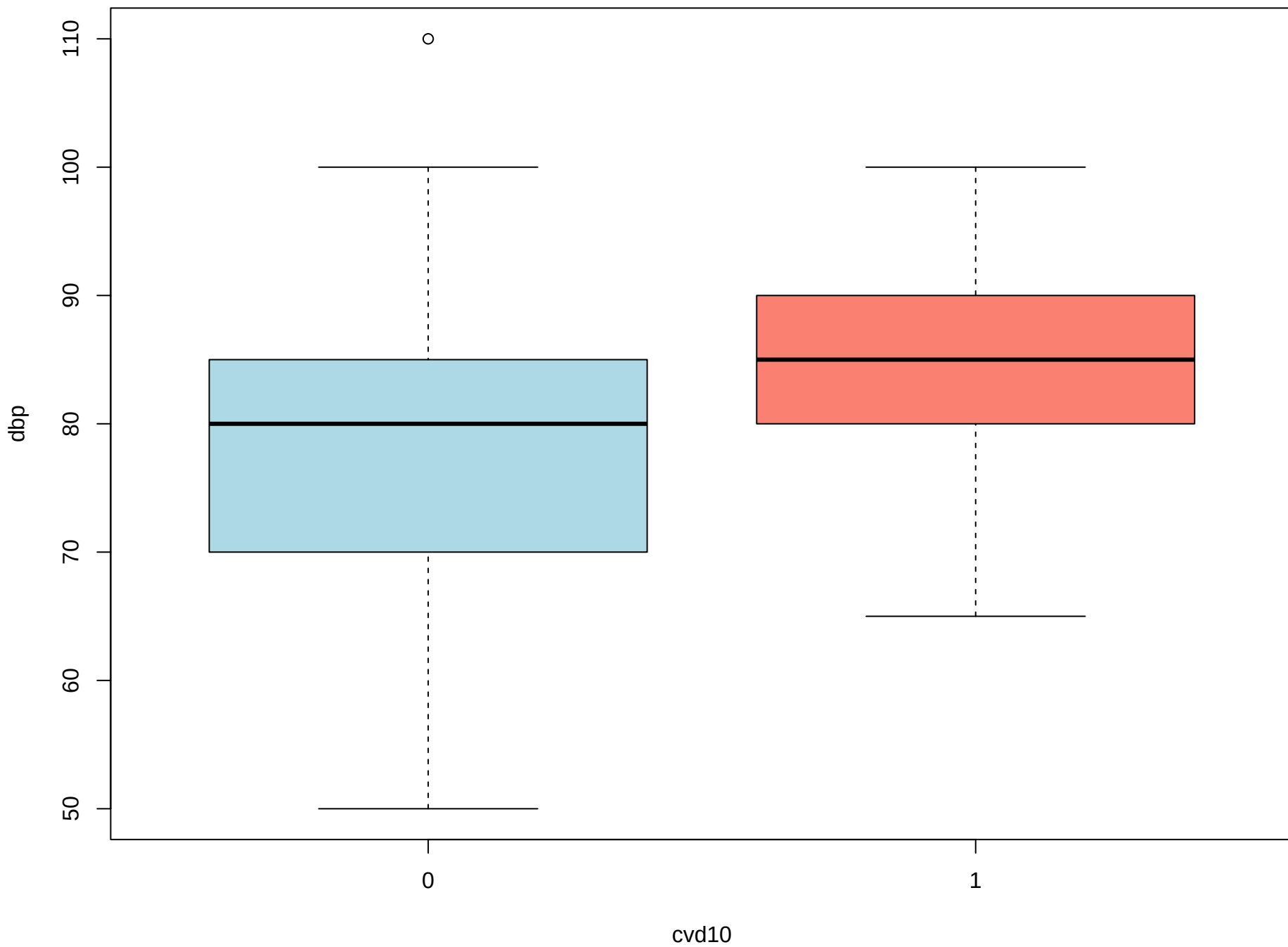
Boxplot: sbp by cvd10



Descriptives: dbp by cvd10

	Mean	SD	Median	Skew
No_CVD	79.3	12.10	80	0.05
Yes_CVD	85.0	9.35	85	-0.27

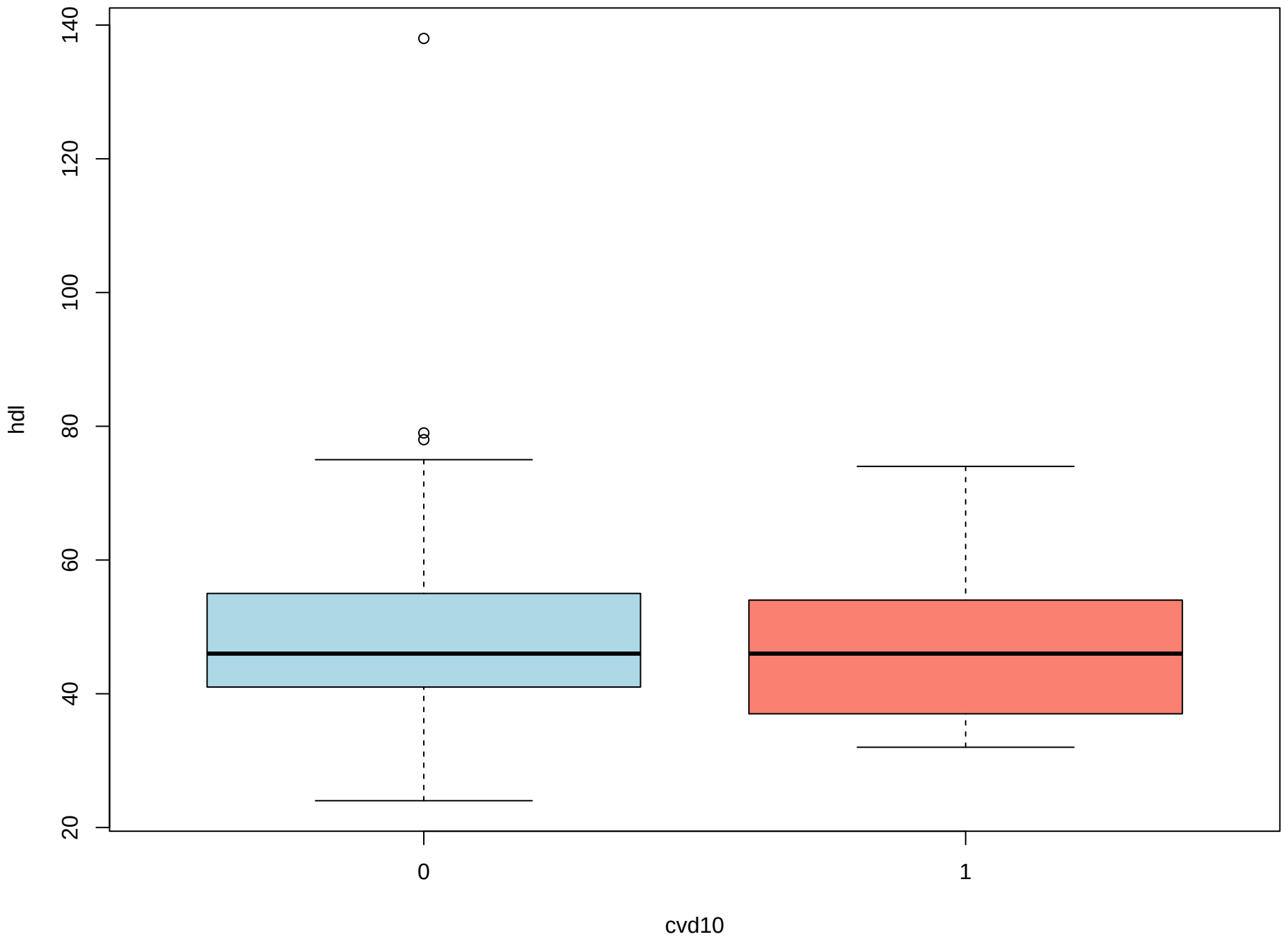
Boxplot: dbp by cvd10



Descriptives: hdl by cvd10

	Mean	SD	Median	Skew
No_CVD	48.05	13.55	46	2.51
Yes_CVD	46.65	11.82	46	0.59

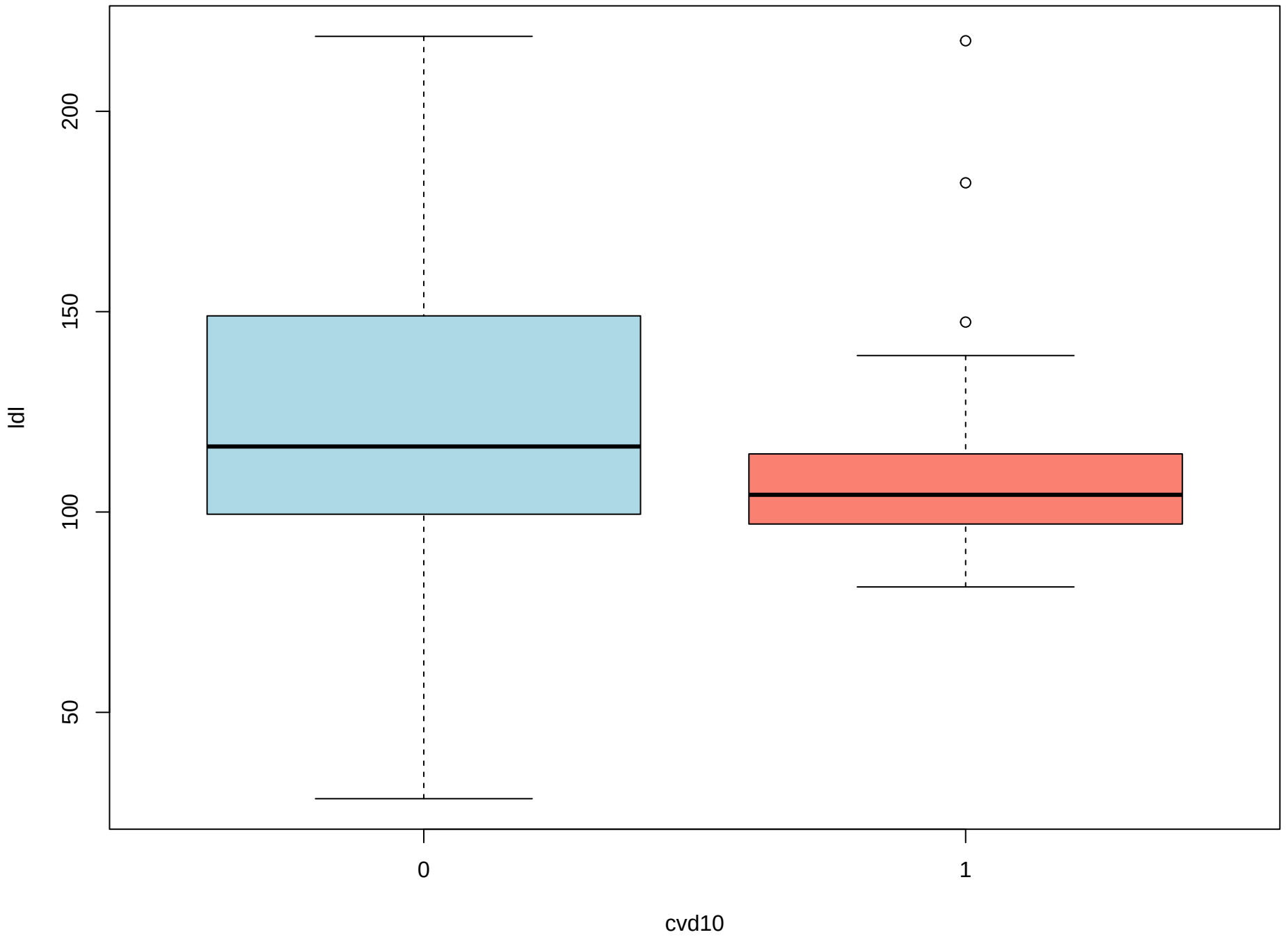
Boxplot: hdl by cvd10



Descriptives: ldl by cvd10

	Mean	SD	Median	Skew
No_CVD	121.47	35.90	116.35	0.18
Yes_CVD	116.35	36.74	104.30	1.42

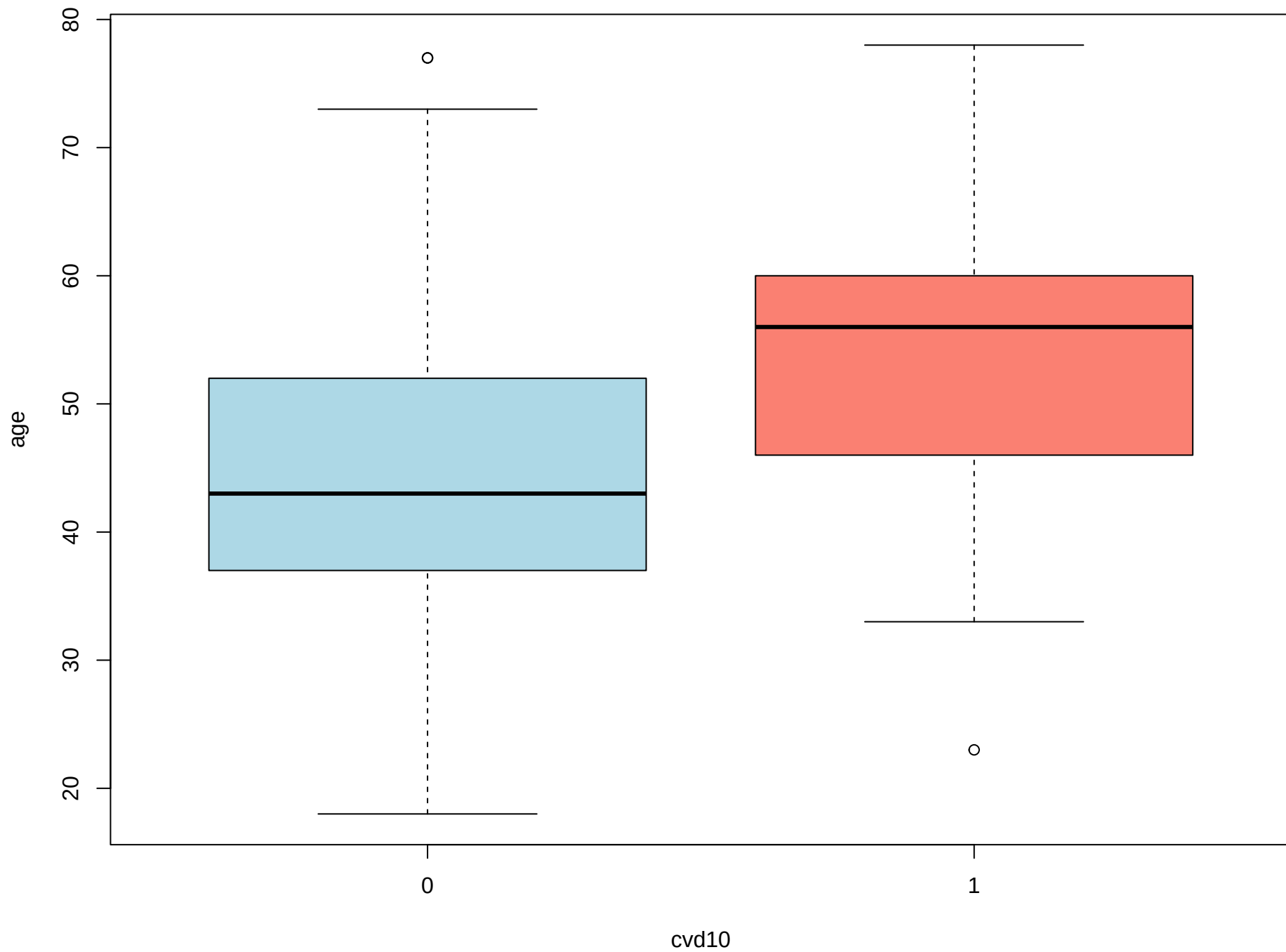
Boxplot: ldl by cvd10



Descriptives: age by cvd10

	Mean	SD	Median	Skew
No_CVD	44.13	12.65	43	0.17
Yes_CVD	53.71	15.98	56	-0.29

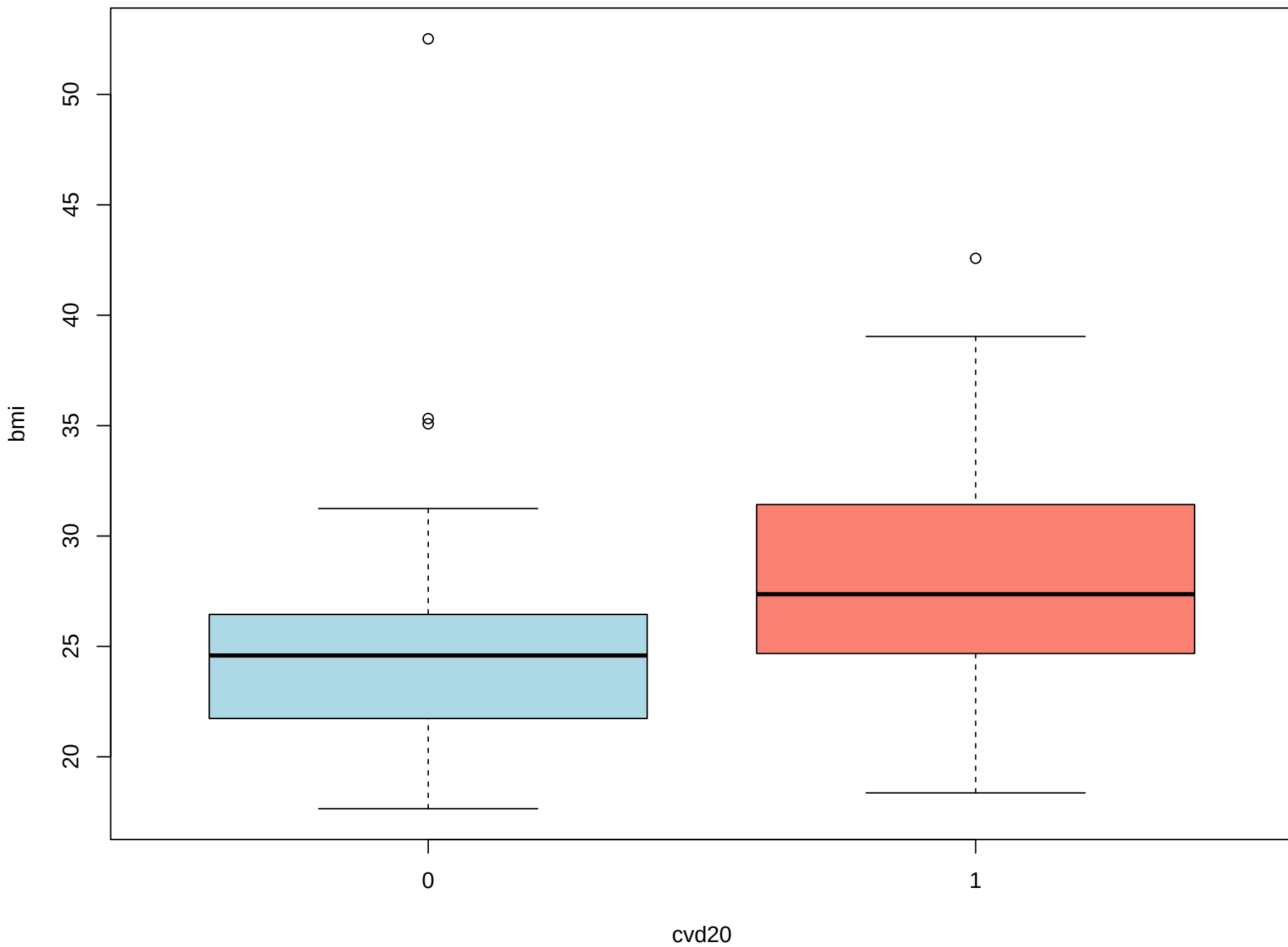
Boxplot: age by cvd10



Descriptives: bmi by cvd20

	Mean	SD	Median	Skew
No_CVD	24.89	4.56	24.59	2.71
Yes_CVD	28.23	4.84	27.36	0.51

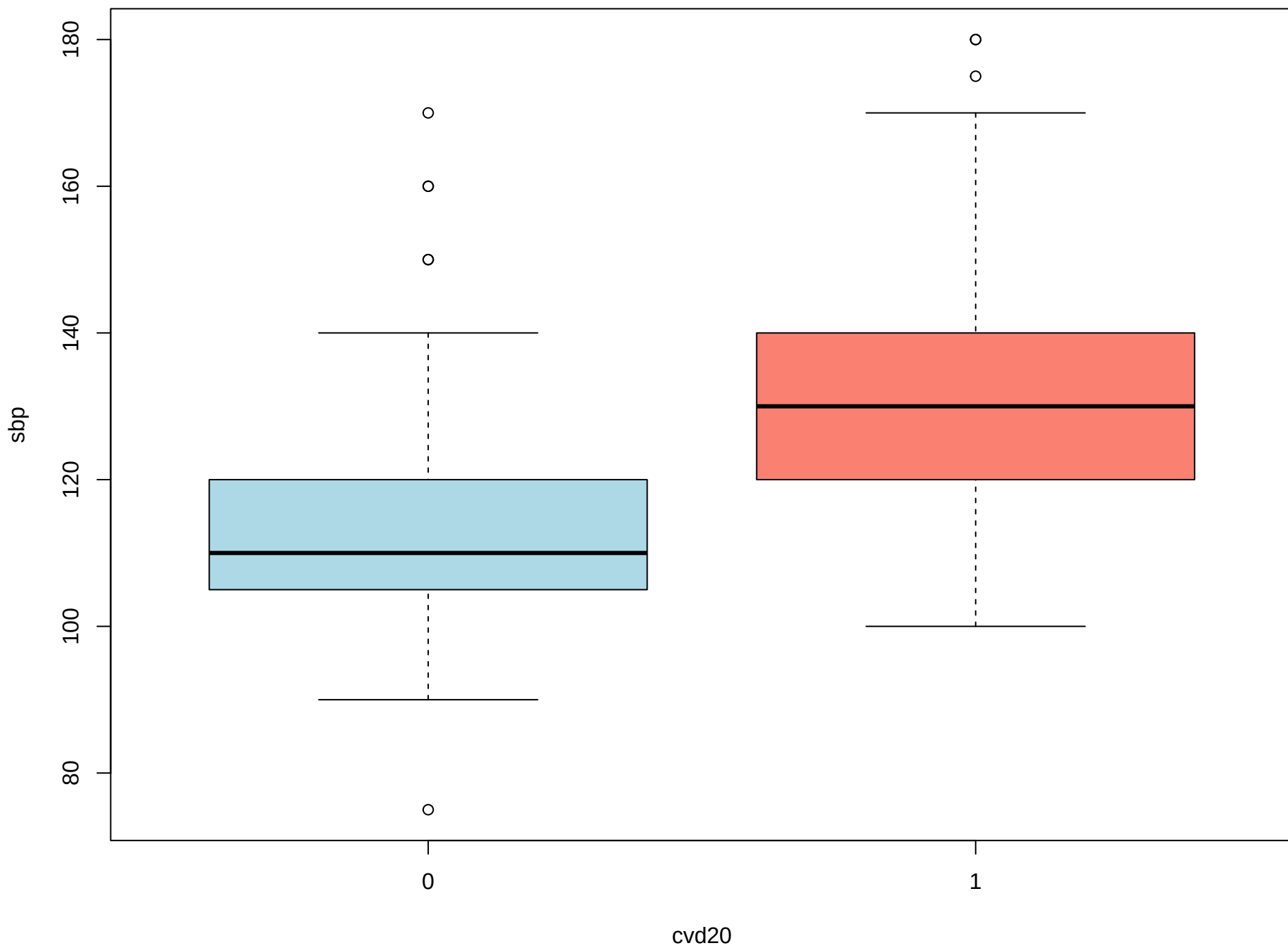
Boxplot: bmi by cvd20



Descriptives: sbp by cvd20

	Mean	SD	Median	Skew
No_CVD	115.58	16.81	110	0.76
Yes_CVD	133.12	18.60	130	0.50

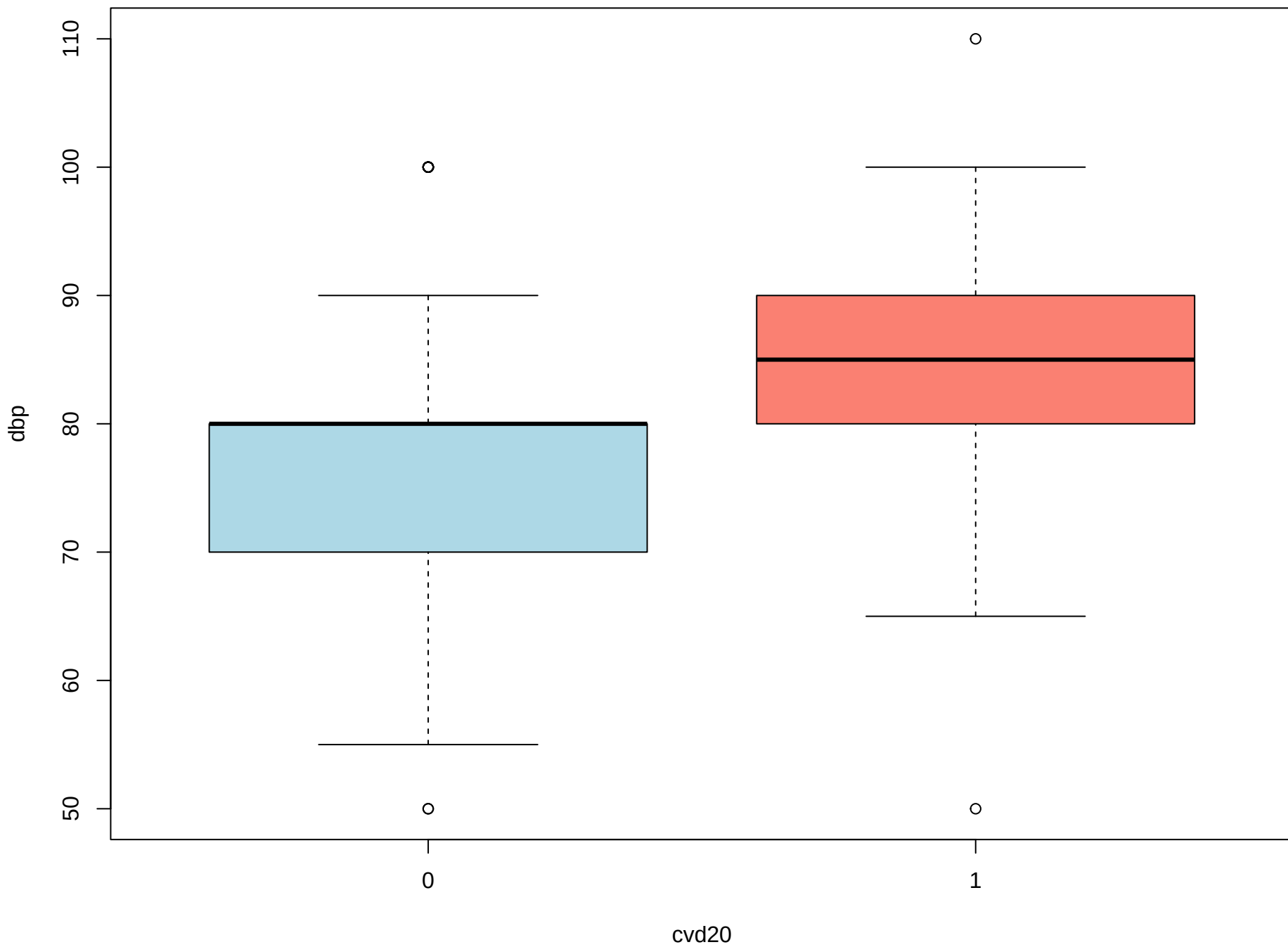
Boxplot: sbp by cvd20



Descriptives: dbp by cvd20

	Mean	SD	Median	Skew
No_CVD	77.09	11.69	80	0.22
Yes_CVD	84.43	10.95	85	-0.35

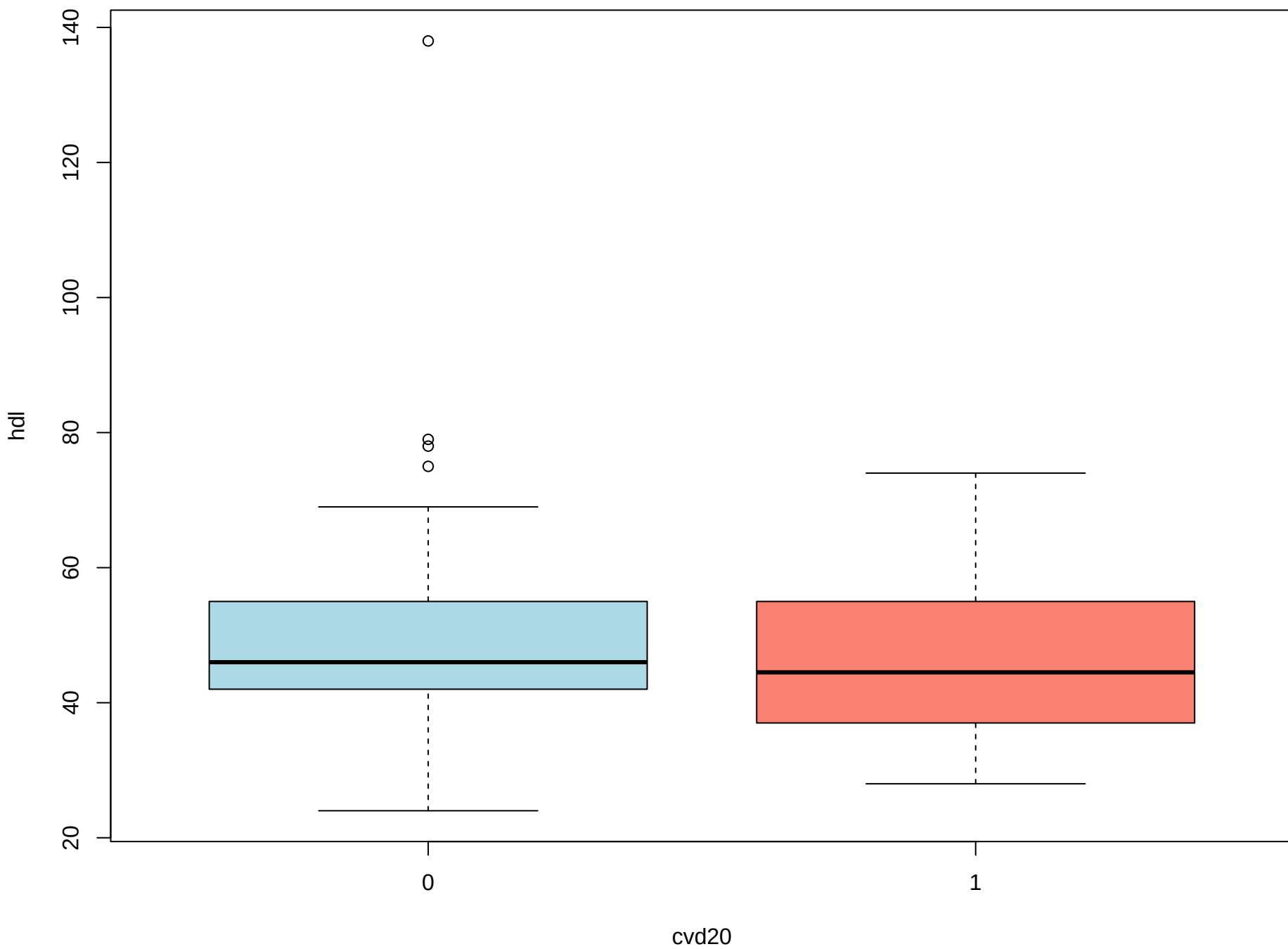
Boxplot: dbp by cvd20



Descriptives: hdl by cvd20

	Mean	SD	Median	Skew
No_CVD	49.17	14.81	46.0	2.65
Yes_CVD	45.89	10.48	44.5	0.36

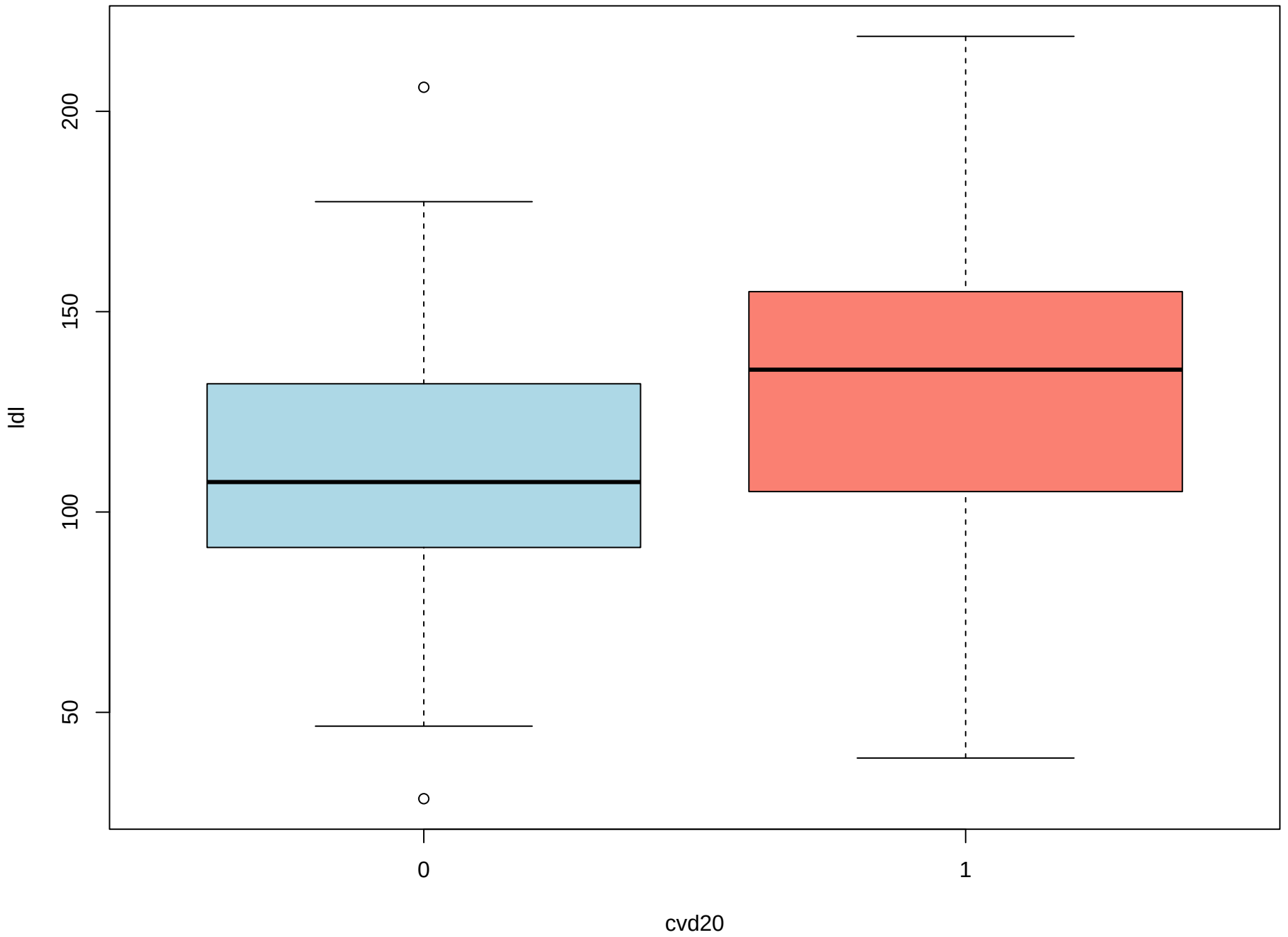
Boxplot: hdl by cvd20



Descriptives: ldl by cvd20

	Mean	SD	Median	Skew
No_CVD	111.85	32.11	107.47	0.31
Yes_CVD	134.68	37.30	135.52	0.17

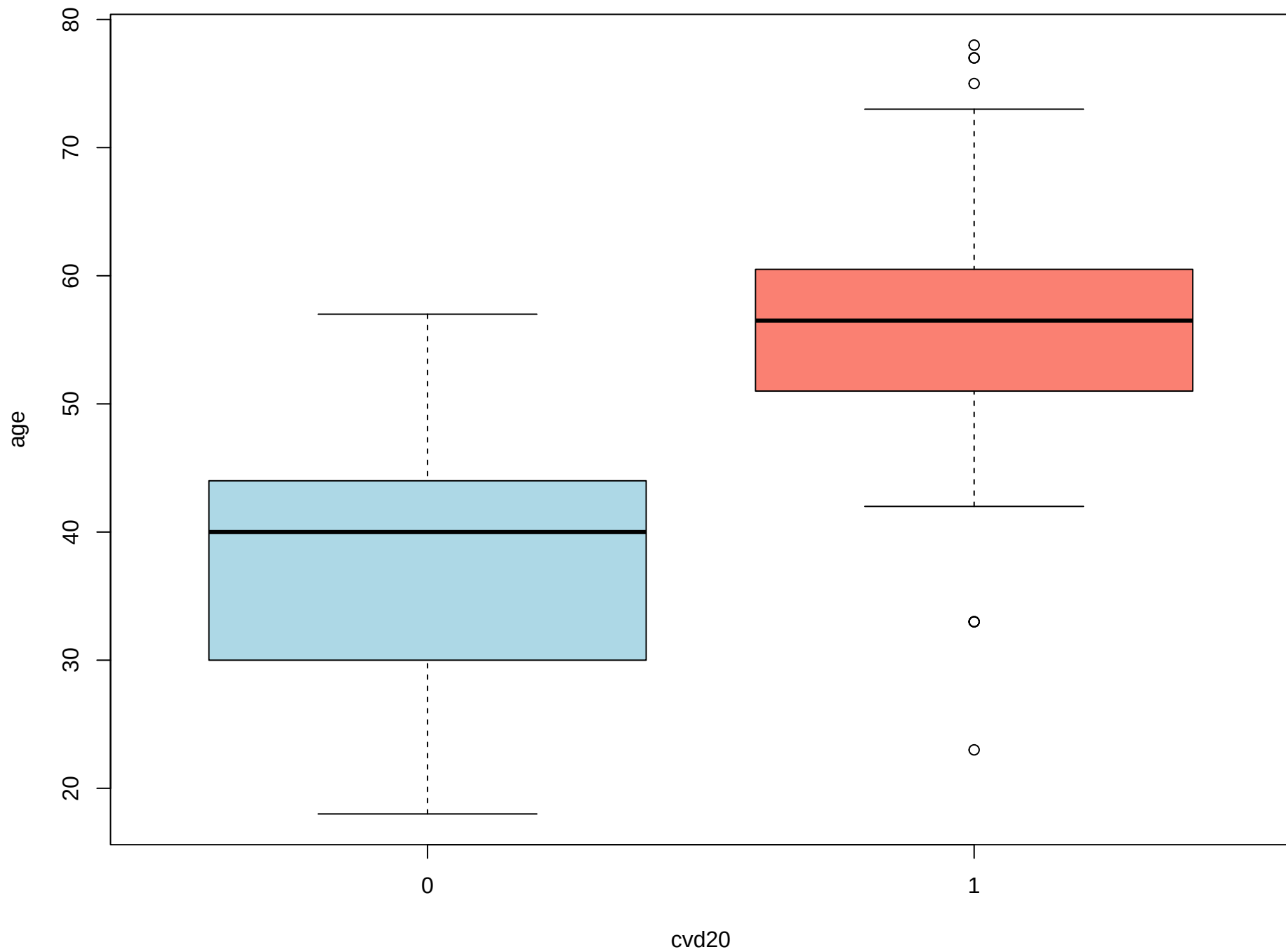
Boxplot: ldl by cvd20



Descriptives: age by cvd20

	Mean	SD	Median	Skew
No_CVD	38.00	9.04	40.0	-0.36
Yes_CVD	56.45	11.12	56.5	-0.41

Boxplot: age by cvd20



Crosstab: htn vs cvd10

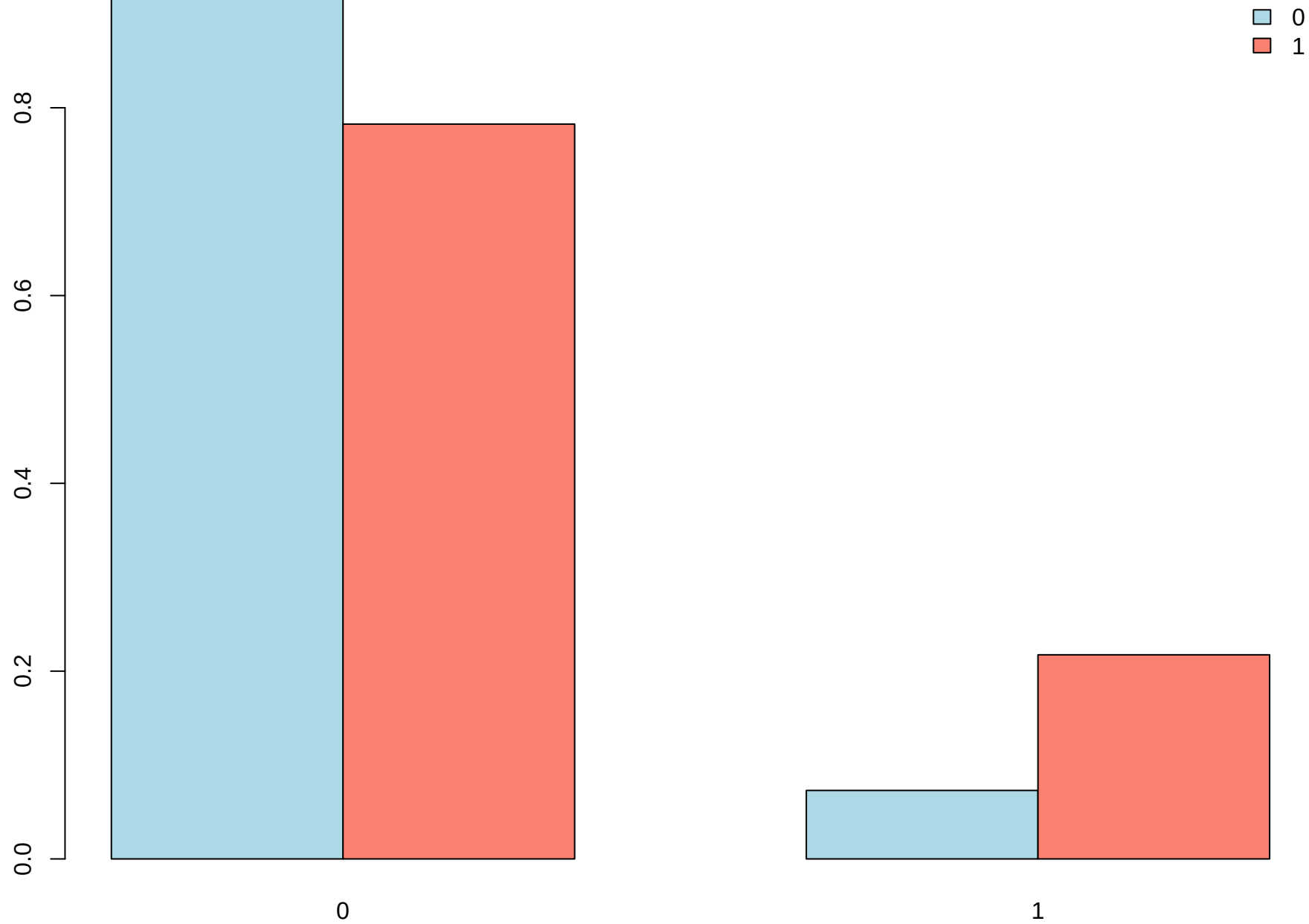
\$Counts

	0	1	Sum
0	89	7	96
1	36	10	46
Sum	125	17	142

\$Row_Percentages

	0	1
0	92.71	7.29
1	78.26	21.74

htn vs cvd10 (Row %)



Crosstab: smoking_ever vs cvd10

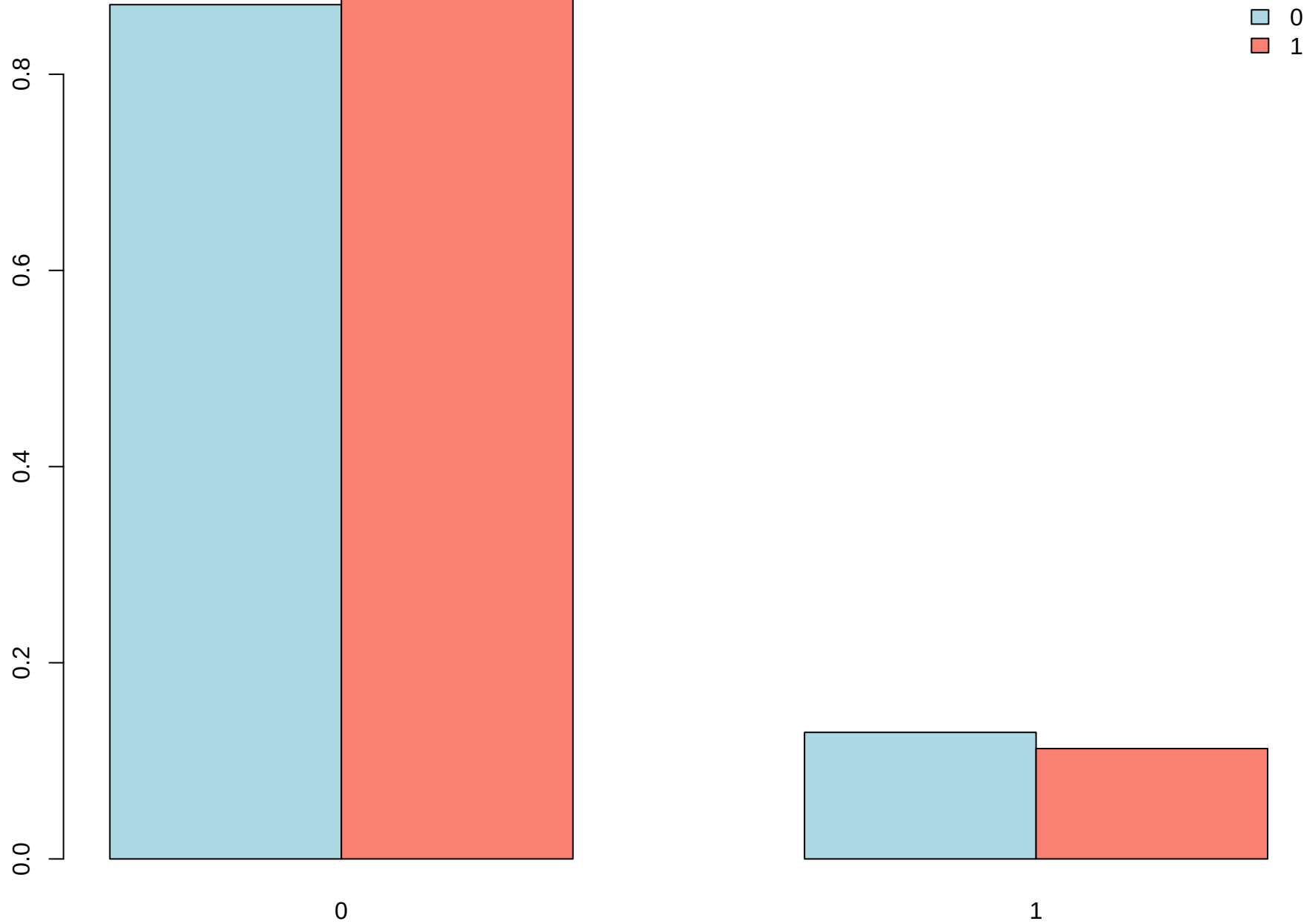
\$Counts

	0	1	Sum
0	54	8	62
1	71	9	80
Sum	125	17	142

\$Row_Percentages

	0	1
0	87.10	12.90
1	88.75	11.25

smoking_ever vs cvd10 (Row %)



Crosstab: physact_level vs cvd10

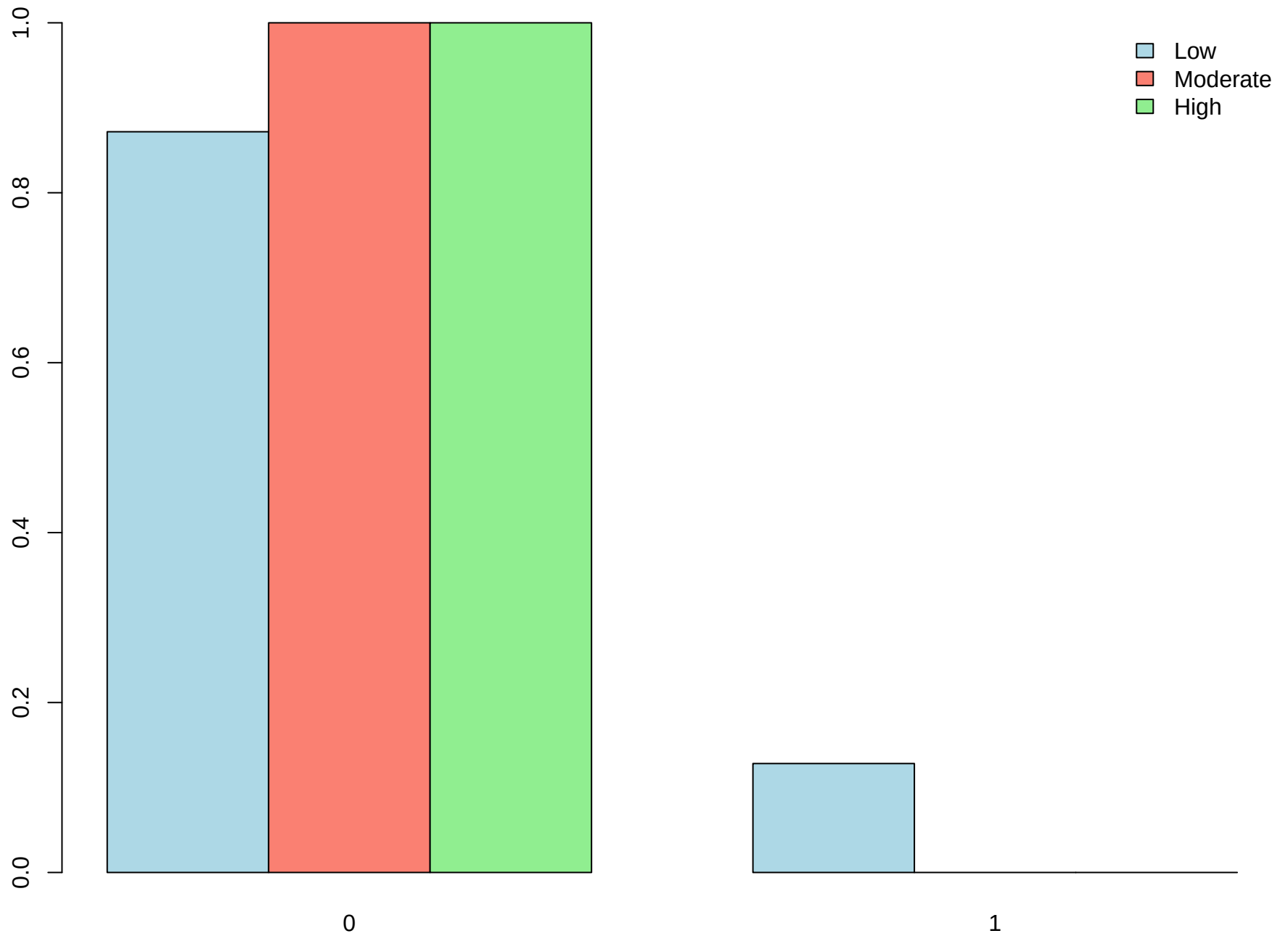
\$Counts

	0	1	Sum
Low	34	5	39
Moderate	5	0	5
High	5	0	5
Sum	44	5	49

\$Row_Percentages

	0	1
Low	87.18	12.82
Moderate	100.00	0.00
High	100.00	0.00

physact_level vs cvd10 (Row %)



Crosstab: htn vs cvd20

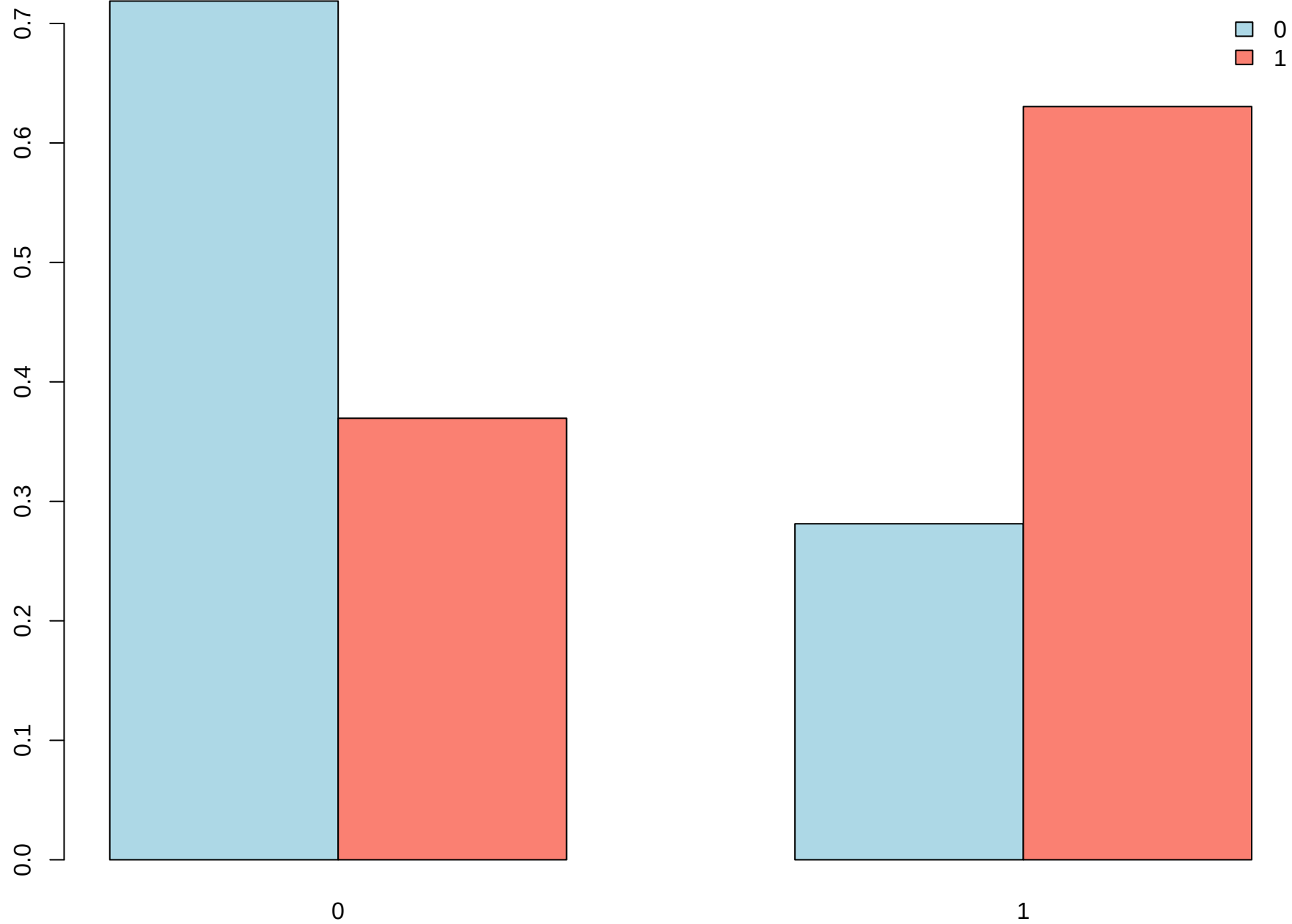
\$Counts

	0	1	Sum
0	69	27	96
1	17	29	46
Sum	86	56	142

\$Row_Percentages

	0	1
0	71.88	28.12
1	36.96	63.04

htn vs cvd20 (Row %)



Crosstab: smoking_ever vs cvd20

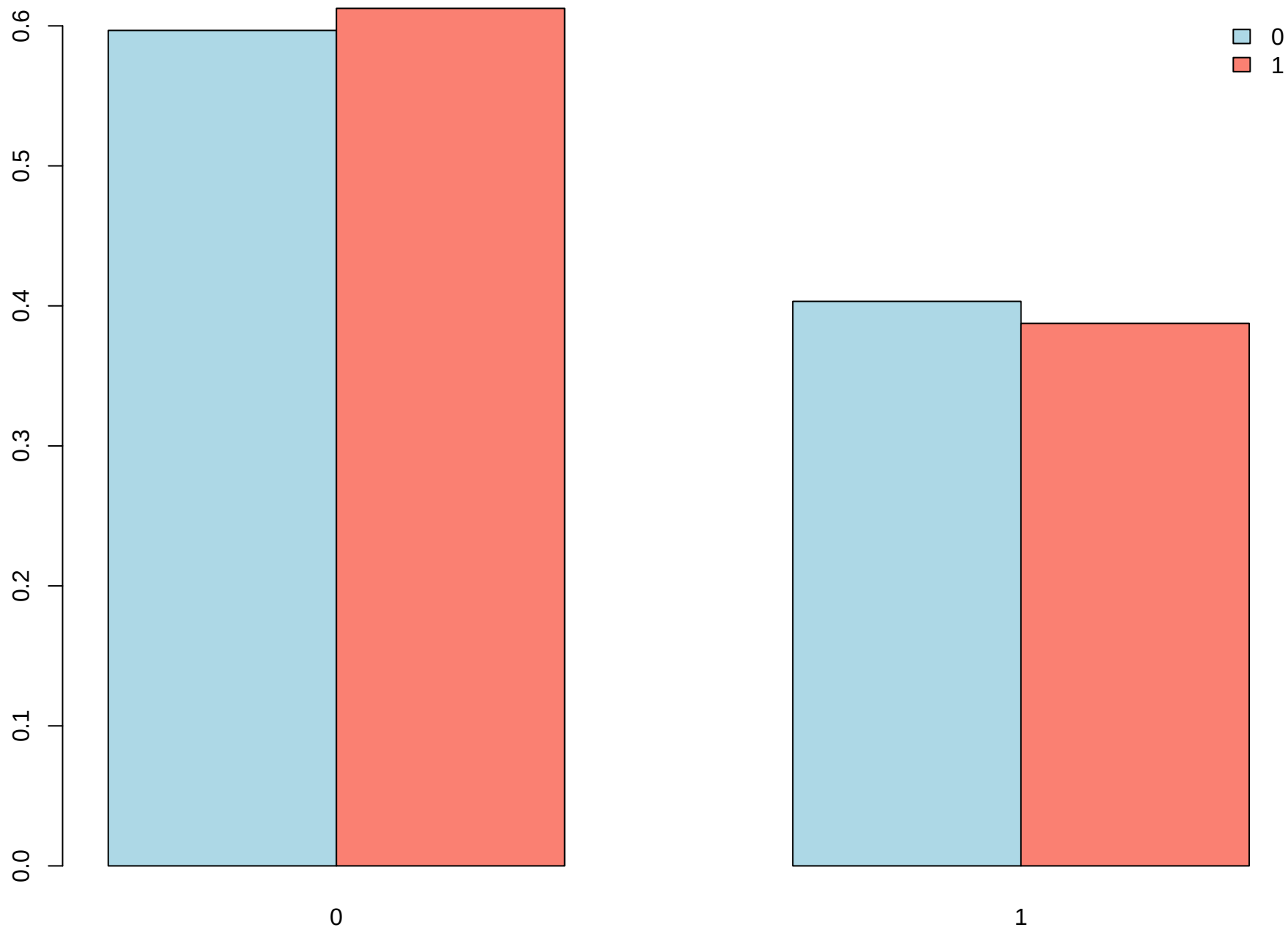
\$Counts

	0	1	Sum
0	37	25	62
1	49	31	80
Sum	86	56	142

\$Row_Percentages

	0	1
0	59.68	40.32
1	61.25	38.75

smoking_ever vs cvd20 (Row %)



Crosstab: physact_level vs cvd20

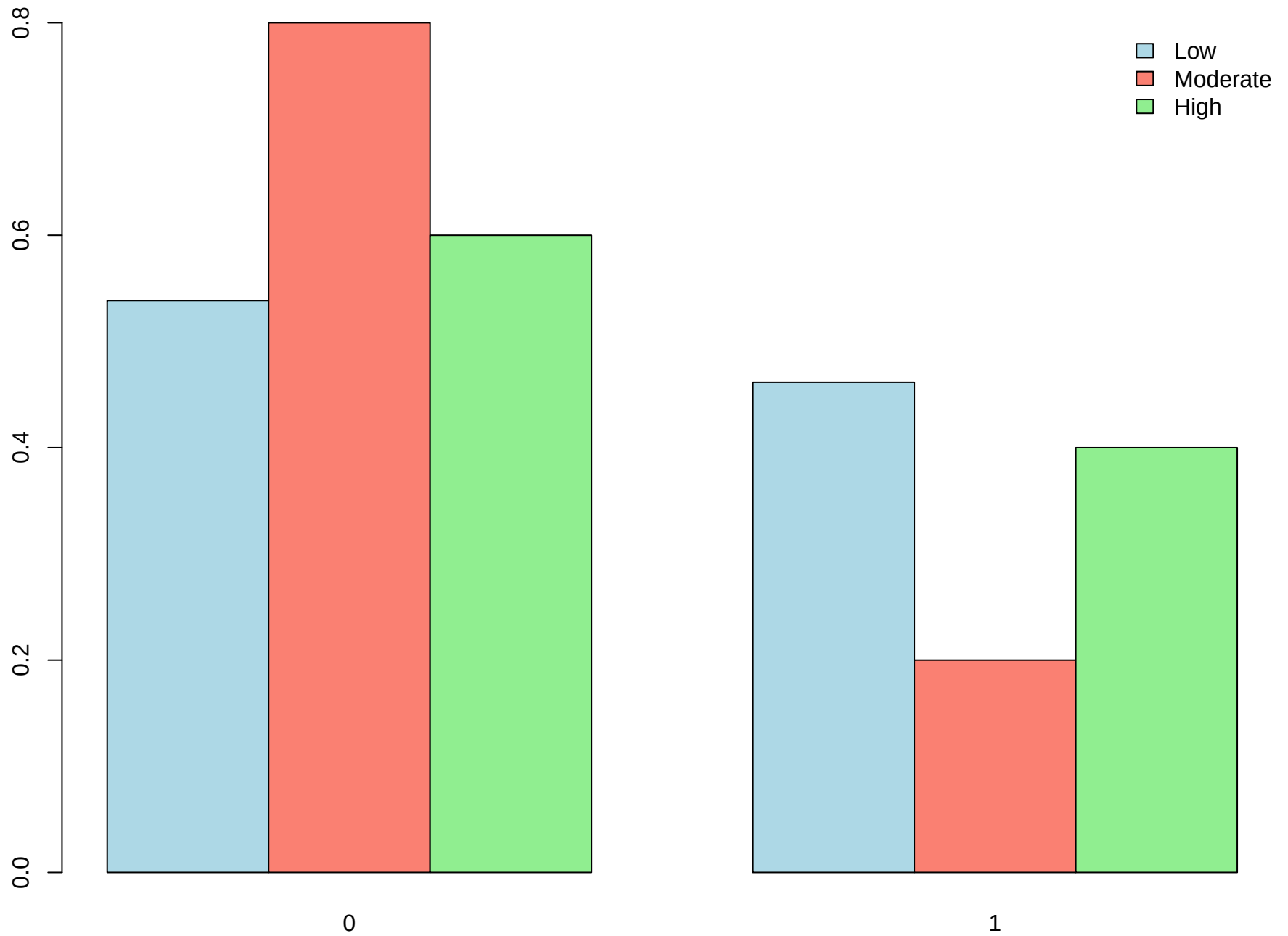
\$Counts

	0	1	Sum
Low	21	18	39
Moderate	4	1	5
High	3	2	5
Sum	28	21	49

\$Row_Percentages

	0	1
Low	53.85	46.15
Moderate	80.00	20.00
High	60.00	40.00

physact_level vs cvd20 (Row %)



Summary of T-tests (P-values)

	p_value
bmi vs cvd10	"0.0024605"
sbp vs cvd10	"0.0052384"
dbp vs cvd10	"0.032771"
hdl vs cvd10	"0.65705"
ldl vs cvd10	"0.59483"
age vs cvd10	"0.028528"
bmi vs cvd20	"7.7405e-05"
sbp vs cvd20	"1.02e-07"
dbp vs cvd20	"0.0002284"
hdl vs cvd20	"0.1246"
ldl vs cvd20	"0.00027755"
age vs cvd20	"< 2.22e-16"

HTN vs cvd10 – Statistical Tests

\$Prop_Test

2-sample test for equality of proportions with continuity correction

data: tbl

X-squared = 4.8648, df = 1, p-value = 0.02741

alternative hypothesis: two.sided

95 percent confidence interval:

-0.001652353 0.290601629

sample estimates:

prop 1 prop 2

0.9270833 0.7826087

\$Chi_Sq

Pearson's Chi-squared test with Yates' continuity correction

data: tbl

X-squared = 4.8648, df = 1, p-value = 0.02741

HTN vs cvd20 – Statistical Tests

\$Prop_Test

2-sample test for equality of proportions with continuity correction

data: tbl

X-squared = 14.448, df = 1, p-value = 0.0001441

alternative hypothesis: two.sided

95 percent confidence interval:

0.1671378 0.5312317

sample estimates:

prop 1 prop 2

0.7187500 0.3695652

\$Chi_Sq

Pearson's Chi-squared test with Yates' continuity correction

data: tbl

X-squared = 14.448, df = 1, p-value = 0.0001441

ANOVA: SBP ~ Physical Activity

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
physact_level	2	265	132.6	0.549	0.581
Residuals	46	11112	241.6		

93 observations deleted due to missingness

Tukey Post-Hoc Test

Tukey multiple comparisons of means
95% family-wise confidence level

Fit: aov(formula = sbp ~ physact_level, data = dataset)

\$physact_level

	diff	lwr	upr	p adj
Moderate-Low	-7.307692	-25.18803	10.57264	0.5869615
High-Low	1.692308	-16.18803	19.57264	0.9714694
High-Moderate	9.000000	-14.80656	32.80656	0.6333305

Multiple Linear Regression (SBP)

Call:

```
lm(formula = f_lm, data = dataset)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-17.7644	-5.5928	0.2282	5.5223	21.4737

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	29.548286	17.460156	1.692	0.0986 .
age	0.038707	0.154354	0.251	0.8033
sexFemale	4.866414	4.115118	1.183	0.2441
bmi	0.624391	0.450504	1.386	0.1736
smoking_ever1	-6.005366	3.349265	-1.793	0.0807 .
physact_levelModerate	7.965709	5.062272	1.574	0.1237
physact_levelHigh	2.143266	5.293345	0.405	0.6878
dbp	0.937043	0.189464	4.946	1.49e-05 ***
hdl	-0.065472	0.148961	-0.440	0.6627
ldl	0.008219	0.041947	0.196	0.8457

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.691 on 39 degrees of freedom

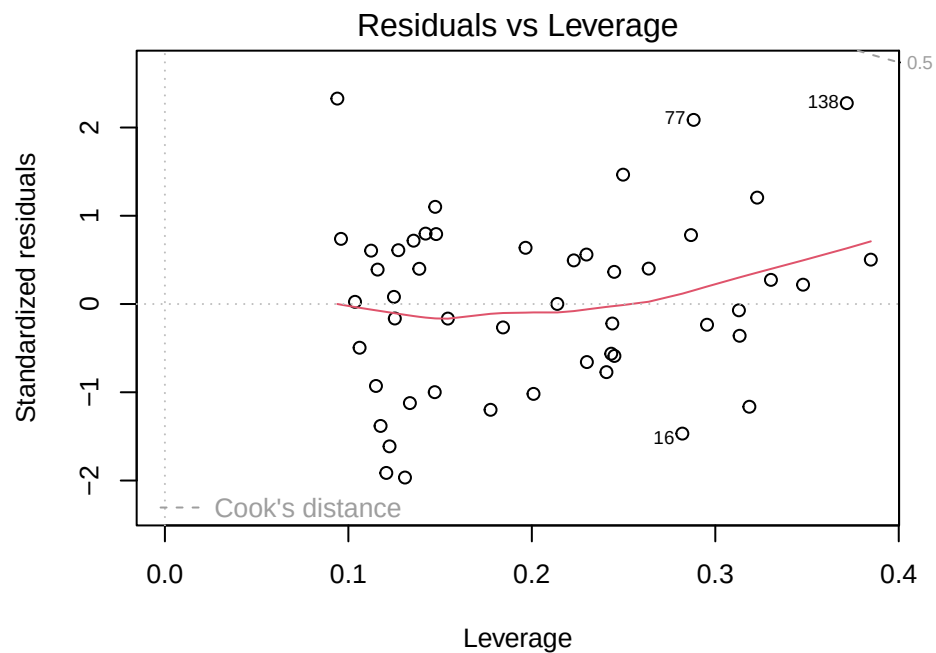
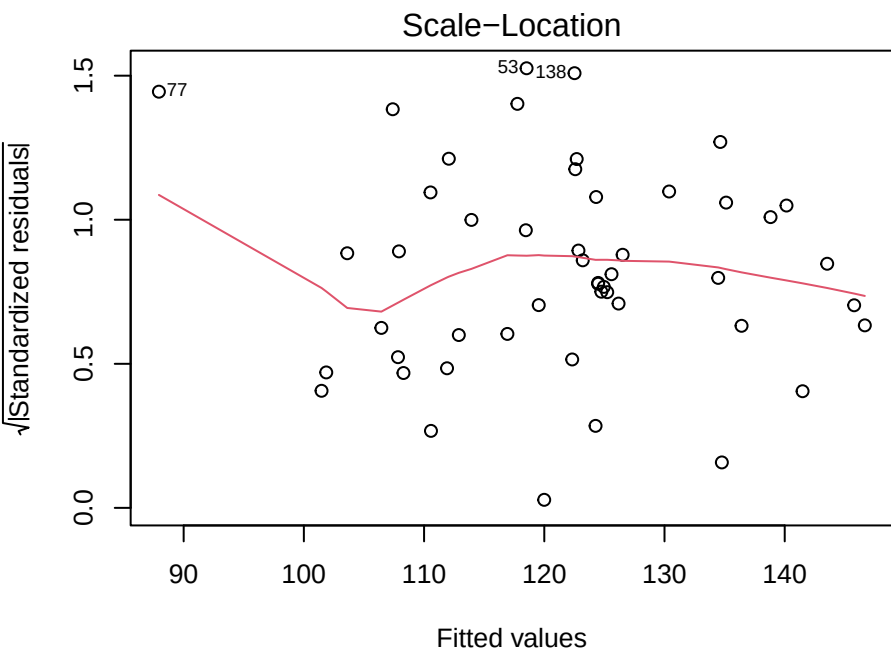
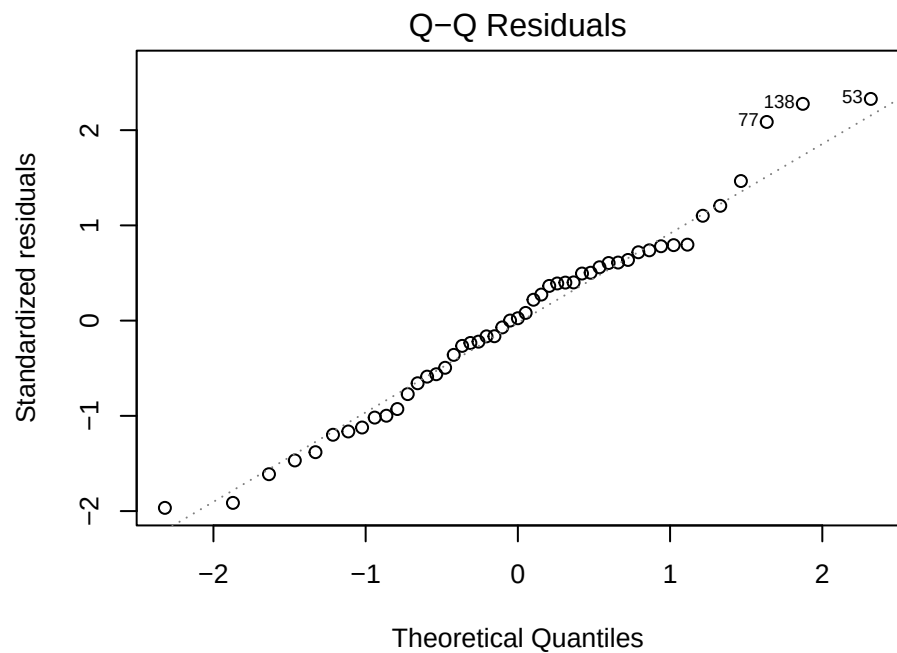
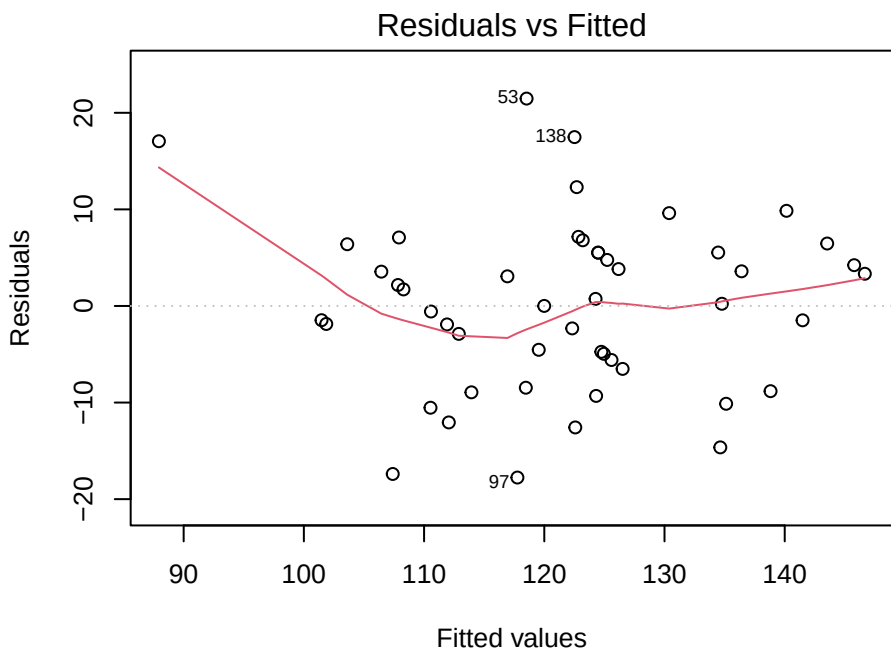
(93 observations deleted due to missingness)

Multiple R-squared: 0.6781, Adjusted R-squared: 0.6038

F-statistic: 9.127 on 9 and 39 DF, p-value: 2.888e-07

VIF Values (Multicollinearity Check)

age	sex	bmi	smoking_ever	physact_level	dbp	hdl
1.77	NA	1.77	NA	NA	1.94	1.52



Logistic Regression: cvd10

Call:

```
glm(formula = f_log, family = binomial, data = dataset)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-7.49598	1.99901	-3.750	0.000177	***
age	0.04222	0.02426	1.741	0.081770	.
sexFemale	-0.00970	0.62211	-0.016	0.987560	
bmi	0.12013	0.05769	2.082	0.037317	*
smoking_ever1	0.09746	0.62562	0.156	0.876208	
htn1	0.11350	0.69195	0.164	0.869707	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

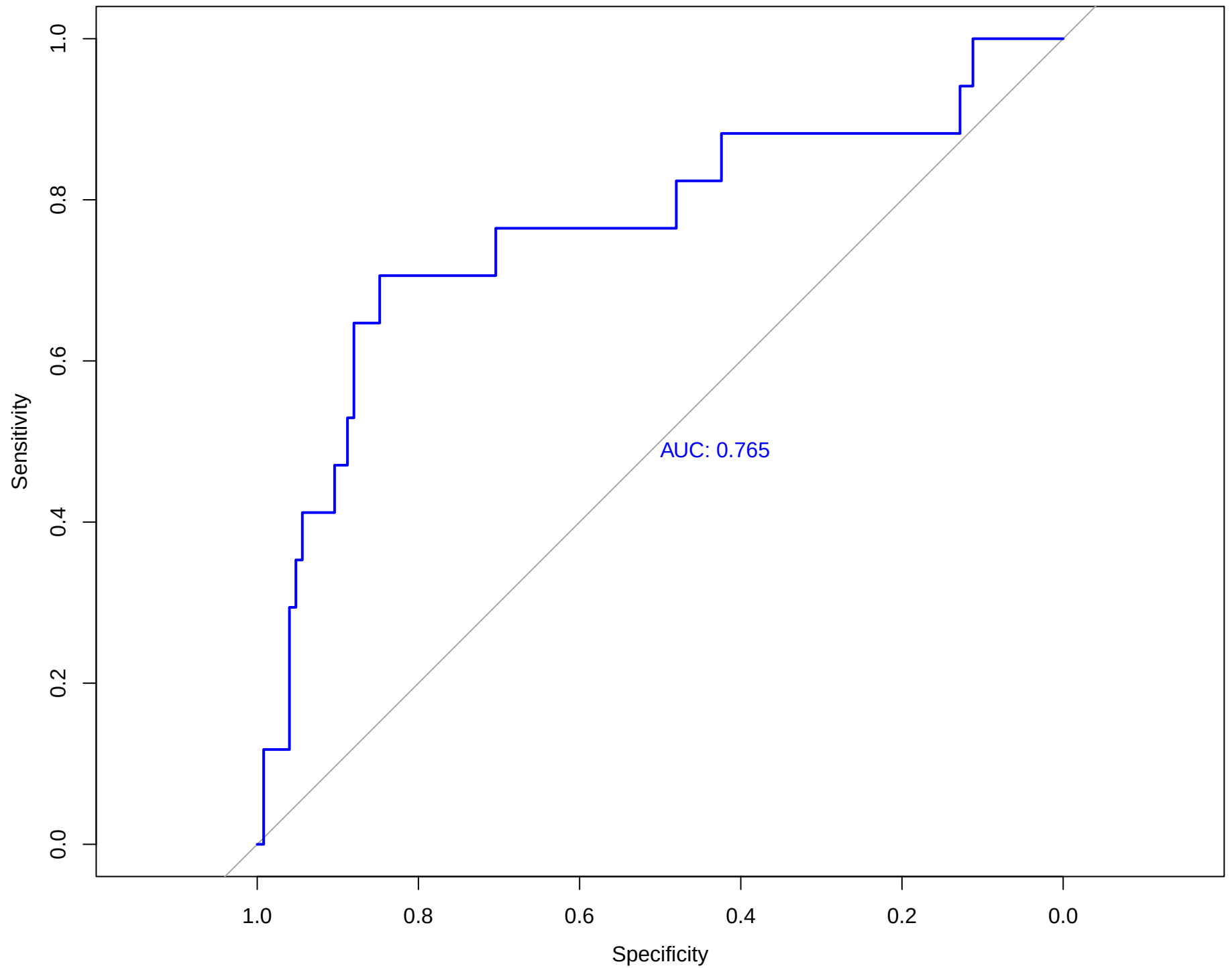
Null deviance: 104.047 on 141 degrees of freedom
Residual deviance: 89.849 on 136 degrees of freedom
AIC: 101.85

Number of Fisher Scoring iterations: 5

Odds Ratios & 95% CI: cvd10

	OR	2.5 %	97.5 %
(Intercept)	0.001	0.000	0.028
age	1.043	0.995	1.094
sexFemale	0.990	0.293	3.352
bmi	1.128	1.007	1.263
smoking_ever1	1.102	0.323	3.757
htn1	1.120	0.289	4.348

ROC Curve: cvd10



Logistic Regression: cvd20

Call:

```
glm(formula = f_log, family = binomial, data = dataset)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-11.58498	2.26484	-5.115	3.14e-07	***
age	0.20685	0.03683	5.616	1.95e-08	***
sexFemale	0.60761	0.53113	1.144	0.253	
bmi	0.03432	0.05647	0.608	0.543	
smoking_ever1	0.14726	0.51753	0.285	0.776	
htn1	0.18954	0.59385	0.319	0.750	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 190.47 on 141 degrees of freedom

Residual deviance: 100.86 on 136 degrees of freedom

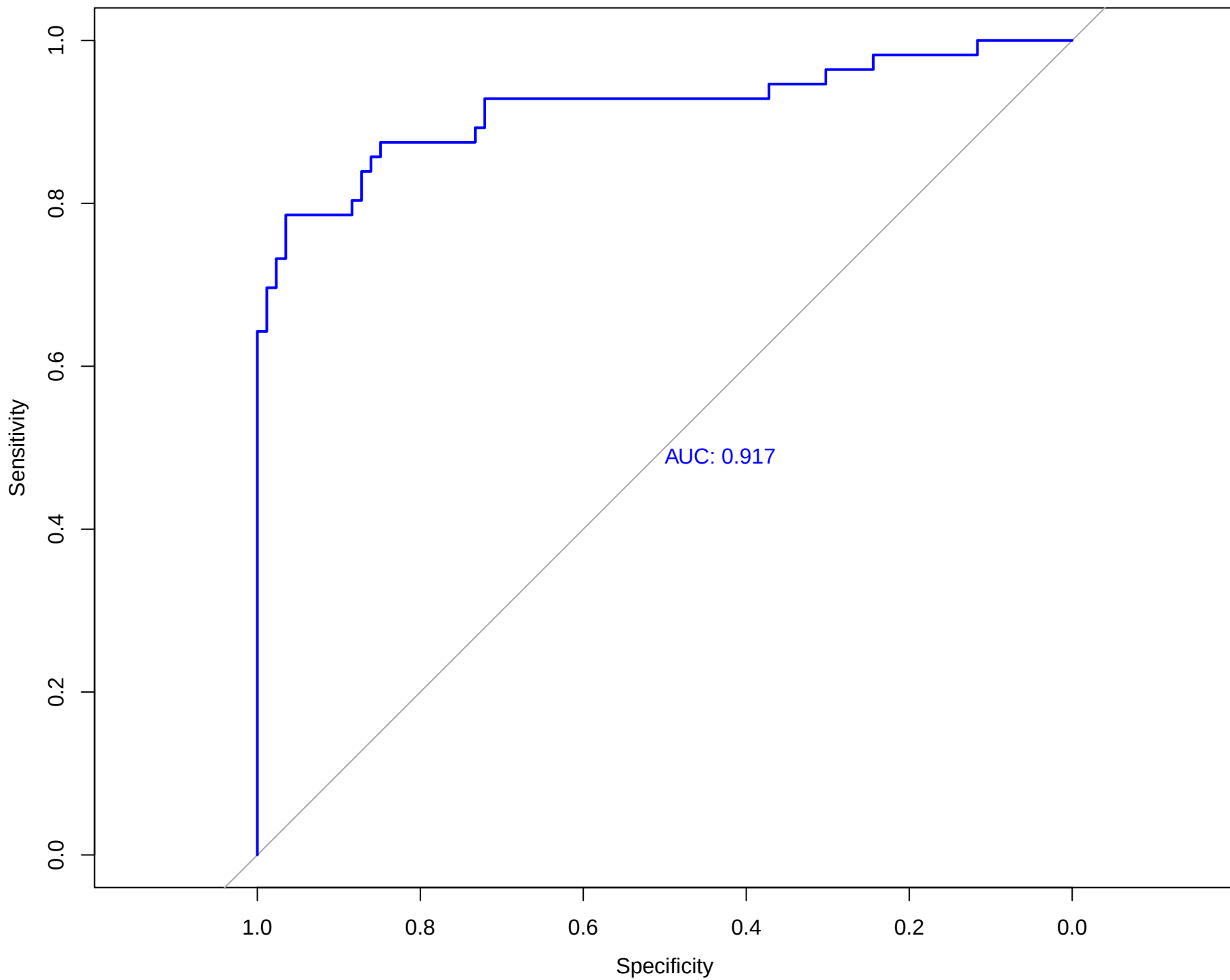
AIC: 112.86

Number of Fisher Scoring iterations: 6

Odds Ratios & 95% CI: cvd20

	OR	2.5 %	97.5 %
(Intercept)	0.000	0.000	0.001
age	1.230	1.144	1.322
sexFemale	1.836	0.648	5.200
bmi	1.035	0.926	1.156
smoking_ever1	1.159	0.420	3.195
htn1	1.209	0.377	3.871

ROC Curve: cvd20



PCA Summary

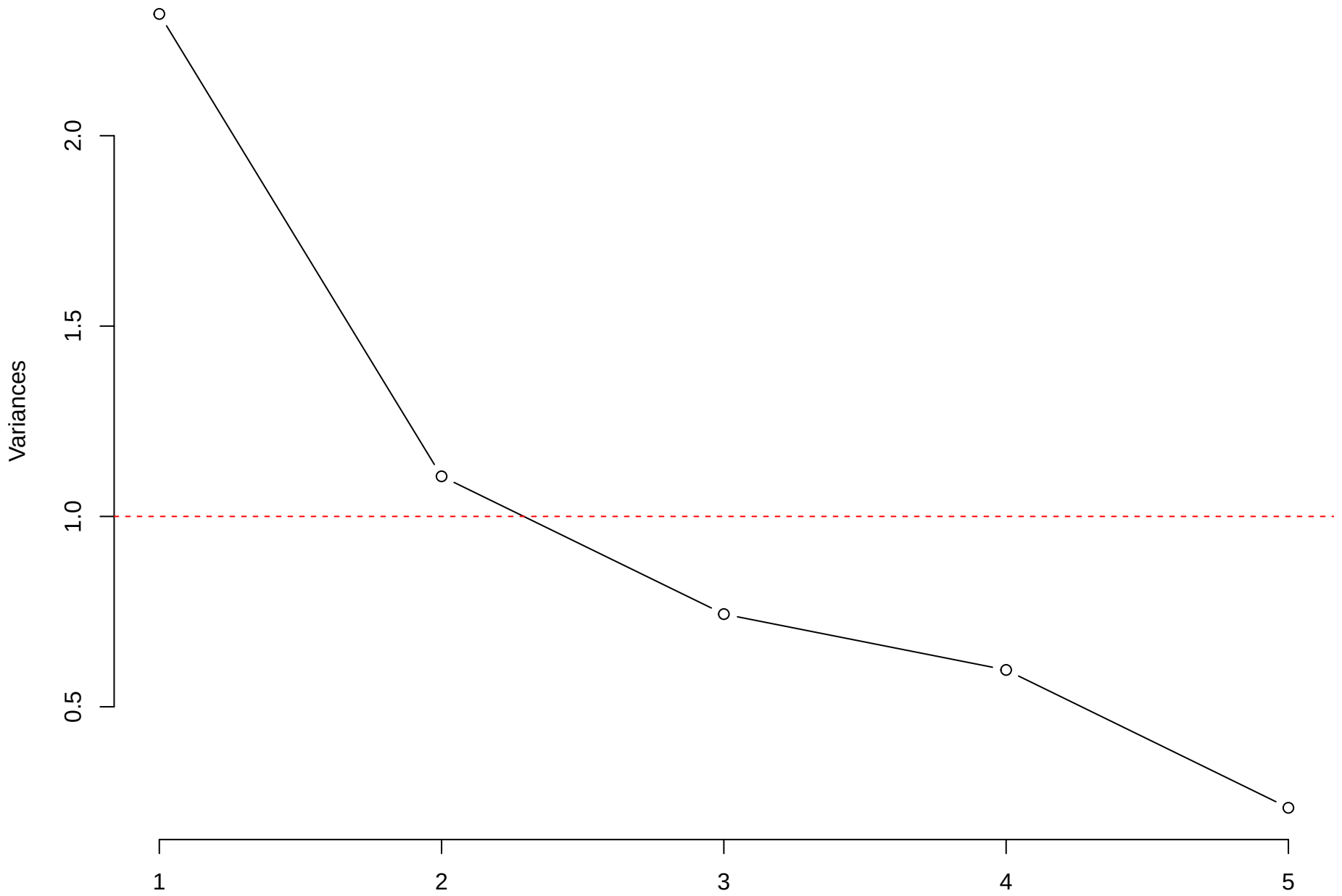
Importance of components:

	PC1	PC2	PC3	PC4	PC5
Standard deviation	1.523	1.0513	0.8622	0.7725	0.48431
Proportion of Variance	0.464	0.2211	0.1487	0.1193	0.04691
Cumulative Proportion	0.464	0.6851	0.8337	0.9531	1.00000

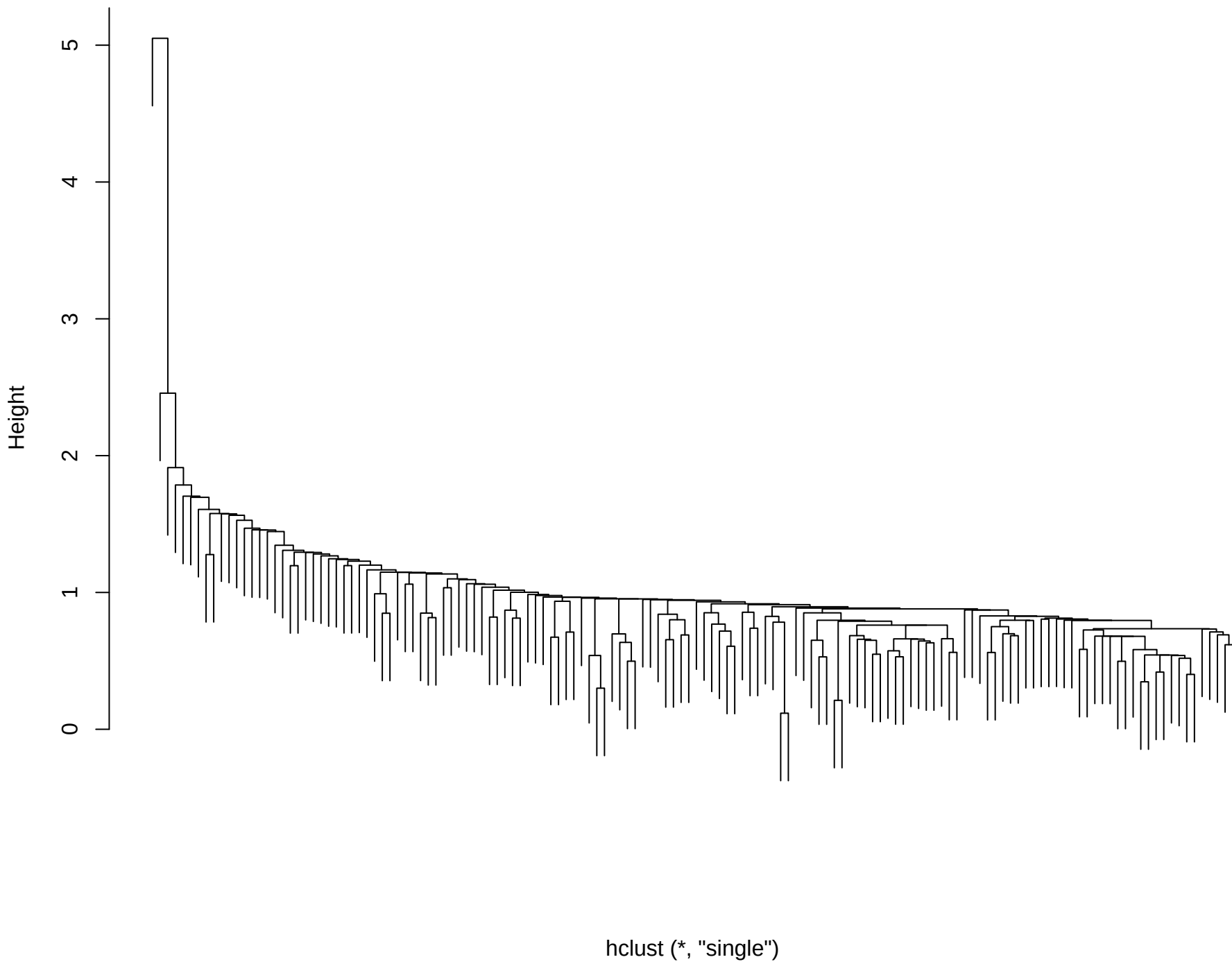
PCA Loadings (First 2 Components)

	PC1	PC2
bmi	0.482	0.086
sbp	0.558	0.266
dbp	0.575	0.218
hdl	-0.296	0.574
ldl	0.197	-0.738

Scree Plot



Dendrogram (Single Linkage)



PCA Factors with Single Linkage Clusters

